PLS_Southwood_FinalProject

2025-04-08

The topic of my project is hurricanes in the US since the 1850s. The data sources is from NOAA at https://www.aoml.noaa.gov/hrd/hurdat/UShurrs_detailed.html. It has reports all huricanes that made landfall in the US since 1851 with time, location, max winds, storm strength, pressure, and more.

My currrent idea for finding am interesting question is to look at the change in storm size over time as well as pressure and wind to see how these variables behave over time. I'm also interested in looking at where these storms take place over time as well.

```
hurricane <- read.csv("Hurricane.csv")
```

```
library(ggmap)
```

```
## Loading required package: ggplot2

## i Google's Terms of Service: <https://mapsplatform.google.com>
##   Stadia Maps' Terms of Service: <https://stadiamaps.com/terms-of-service/>
##   OpenStreetMap's Tile Usage Policy: <https://operations.osmfoundation.org/policies/tiles/>
## i Please cite ggmap if you use it! Use `citation("ggmap")` for details.
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##     filter, lag
##
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

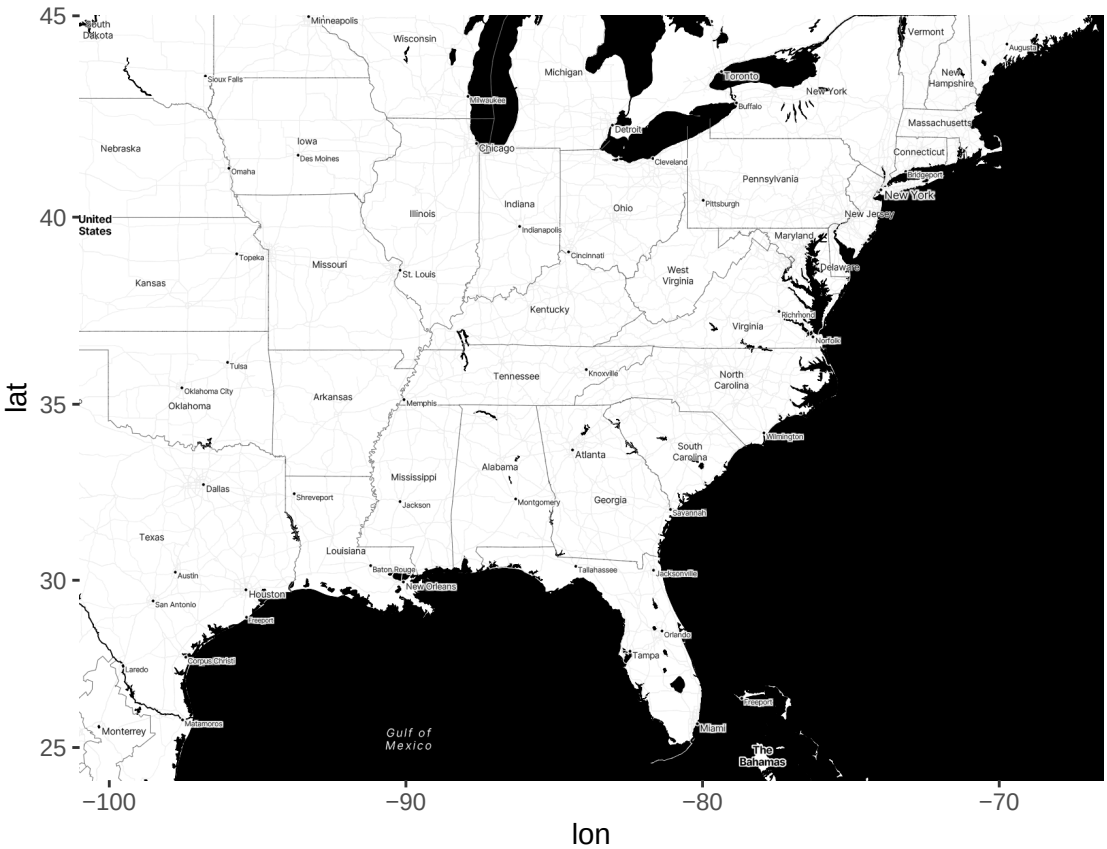
```
# This is my API key--if you do a lot of mapping on your
# own, you'll need to make your own key.
# Instructions:
# https://docs.stadiamaps.com/guides/migrating-from-stamen-map-tiles/#ggmap
# The cache=TRUE argument above means that R will
# save the contents of this chunk so it doesn't download
# the map each time you knit.
api_key <- "7c8c56fe-3e81-4259-b812-579db266b926"
ggmap::register_stadiamaps(api_key)

east_coast <- c(left = -101, bottom = 24, right = -66.5, top = 45)

map <- get_stadiamap(east_coast, maptype = "stamen_toner", zoom = 6)
```

```
## i © Stadia Maps © Stamen Design © OpenMapTiles © OpenStreetMap contributors.
```

```r
ggmap(map)
```



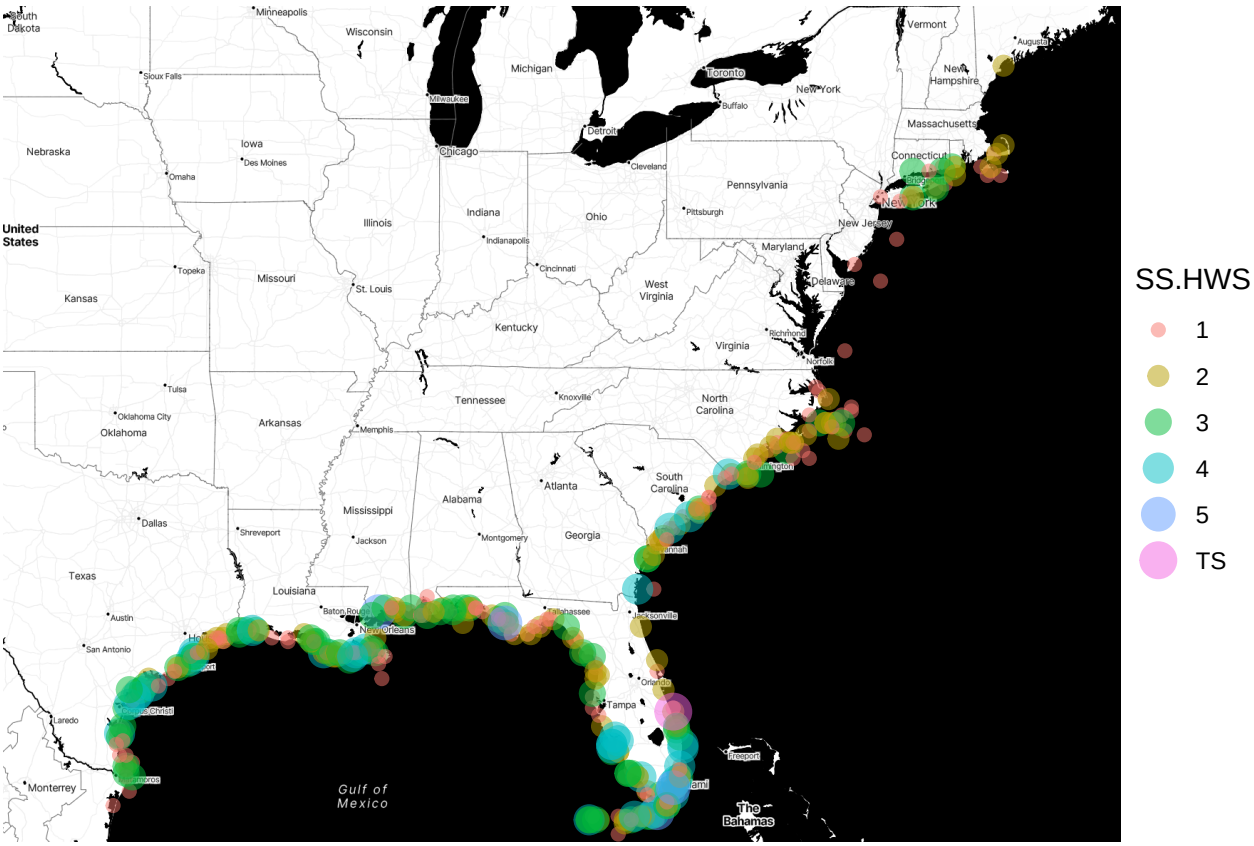

```r
hurricane$Longitude <- as.numeric(sub("W", "", hurricane$Longitude)) * -1
hurricane$Latitude <- as.numeric(sub("N", "", hurricane$Latitude))
```

```r
library(ggplot2)

ggmap(map) +
  geom_point(data = hurricane, aes(x = Longitude, y = Latitude, size = SS.HWS , color = SS.HWS),
             alpha = 0.5) +
  theme_void()
```

```
## Warning: Using size for a discrete variable is not advised.

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
hurricane$Central.Pressure..mb. <- as.numeric(hurricane$Central.Pressure..mb.)
```

```
## Warning: NAs introduced by coercion
```

```
hurricane$Central.Pressure..mb. <- abs(as.numeric(hurricane$Central.Pressure..mb.))

hurricane <- hurricane %>%
  mutate(
    Central.Pressure..mb. = abs(as.numeric(Central.Pressure..mb.)),
    SS.HWS = as.factor(SS.HWS)
  ) %>%
  filter(SS.HWS != "TS")


plot1 <- ggplot(hurricane, aes(x = Central.Pressure..mb., y = Max.Winds...kt., color = SS.HWS)) +
 geom_point() +
  theme_minimal() +
  labs(title = "Lower Pressure Leads to Stronger Winds and Powerful Storms",
       x = "Central Pressure",
       y = "Maximum Winds",
       color = "Storm Category"
       )  +
  geom_smooth(se = TRUE, color = "darkgrey")
```

states bar graph

```r
library(dplyr)
library(tidyr)

library(dplyr)
library(ggplot2)
library(tidyr)

plot2 <- hurricane %>%
  mutate(States_Affected = strsplit(States..Affected, ",")) %>%
  unnest(States_Affected) %>%
  mutate(States_Affected = trimws(States_Affected)) %>%
  mutate(States_Affected = case_when(
    grepl("ATX|BTX|CTX|TX1|CX4", States_Affected) ~ "Texas",
    grepl("LA", States_Affected) ~ "Louisiana",
    grepl("MS", States_Affected) ~ "Mississippi",
    grepl("AL", States_Affected) ~ "Alabama",
    grepl("AFL|BFL|CFL|DFL|CF3", States_Affected) ~ "Florida",
    grepl("GA", States_Affected) ~ "Georgia",
    grepl("SC", States_Affected) ~ "South Carolina",
    grepl("NC", States_Affected) ~ "North Carolina",
    grepl("NY", States_Affected) ~ "New York",
    grepl("MA", States_Affected) ~ "Massachusetts",
    TRUE ~ States_Affected)) %>%
  filter(States_Affected %in% c("Florida", "Texas", "Louisiana", "North Carolina",
                                "South Carolina", "Alabama", "Georgia",
                                "Mississippi", "New York", "Massachusetts")) %>%
  count(States_Affected) %>%


  ggplot(aes(x = reorder(States_Affected, -n), y = n, fill = States_Affected)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  labs(
    title = "Florida and Texas Lead the United States in Hurricanes since 1851",
    x = "",
    y = "Number of Storms") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none")
```

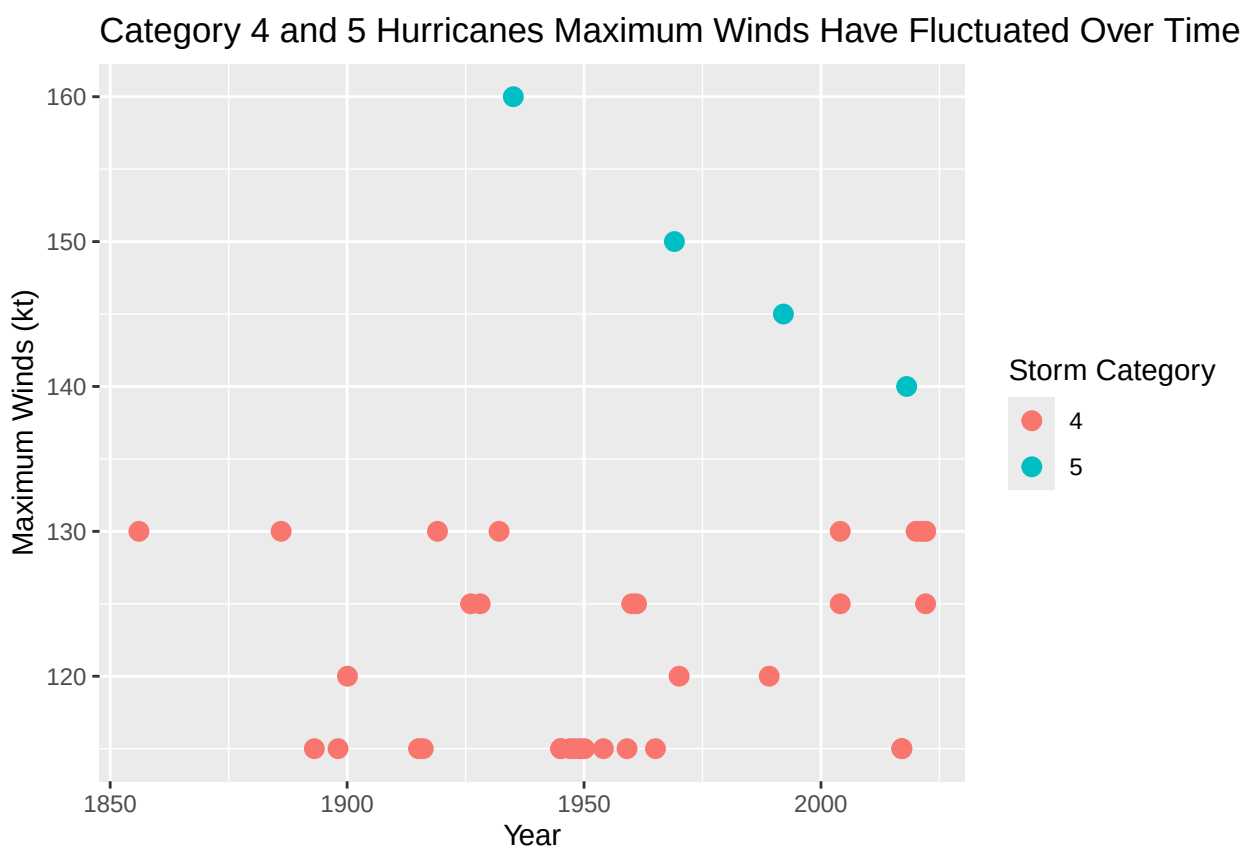
top states plot pressure over time or wind over time

```r
hurricane_filtered <- hurricane %>%
  mutate(Date = as.Date(Date, format = "%m/%d/%Y"),
         Year = as.numeric(format(Date, "%Y")),
         DateNum = as.numeric(format(Date, "%Y%m%d"))) %>%
  filter(SS.HWS %in% c(4, 5))

ggplot(hurricane_filtered, aes(x = DateNum, y = Max.Winds...kt., color = SS.HWS)) +
  geom_point(size = 3) +
  labs(
    title = "Category 4 and 5 Hurricanes Maximum Winds Have Fluctuated Over Time ",
    x = "Year",
    y = "Maximum Winds (kt)",
     color = "Storm Category"
```

```
  ) +
  scale_x_continuous(
    breaks = c(18500101, 19000101, 19500101, 20000101),
    labels = c("1850", "1900", "1950", "2000")


  )
```




```
ggsave("thirdplot.png", width = 7, height = 5, dpi = 300)
```

```
# plot1 + plot2


# ggsave("twoplots.png", width = 14, height = 7, dpi = 300)
```

Extra credit: Yes I would like to make my project available to be shared in future classes and to the authors of the CCC for extra credit. Thank you!

LLMs: I used LLMs in my project to help me with debugging issues that arrose during plotting. It helped me realize I was trying to plot "date" instead of "Date". Whoops!